

LOCAL POLYNOMIAL PERIODOGRAM FOR TIME-VARYING FREQUENCY ESTIMATION

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Summary: The local polynomial time-frequency transform (LPTFT) and the local polynomial time-frequency periodogram (LPP) are proposed in order to estimate a rapidly time-varying frequency $\Omega(t)$ of a harmonic signal. The LPTFT gives a time-frequency energy distribution over the $t - (\Omega(t), d\Omega(t)/dt, \dots, d^{m-1}\Omega(t)/dt^{m-1})$ space, where m is the degree of the LPTFT. The LPTFT enables one to estimate both the time-varying frequency and its derivatives. The technique is based on the fitting of a local polynomial approximation of the frequency which implements a high-order nonparametric regression. The a priori information about bounds for the frequency and its derivatives can be incorporated to improve the accuracy of the estimation. The estimator is shown to be strongly consistent and Gaussian for a polynomial frequency. The asymptotic MSE of the estimators of $d^s\Omega(t)/dt^s$, $s = 0, 1, 2, \dots, m-1$, are obtained for the case of a frequency with bounded m -derivative. Simulation results are presented.

1. INTRODUCTION

The estimation of frequencies of sinusoids in additive noise is one of the very important problems in the theory and in the application of time-series analysis. A large number of recently published papers are devoted to the

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construction and analysis of frequency estimation methods, starting from different assumptions and using different methodologies (Kay, 1988, Haykin, 1991). An important class of such algorithms is based on the parametric representation of signals with an estimation of amplitude, phase and frequency. It has been shown, for example, that the generalized least squares, maximum likelihood, and iterative filtering methods can be successfully applied to the estimation even in the case of low signal-to-noise ratio and small data sets.

The least squares (LS) method and periodogram analysis are supported by strong mathematical results. In particular, Walker (1971) and Hannan (1973) proved that the LS method and periodogram analyses give estimates which converge with the rate $O(N^{-3/2})$, where N is the number of observations.

The time-varying frequency is a character feature of a wide range of signals, including speech, music, and other acoustic, biological and geophysical signals. A current review of the field is given by Boashash (1992).

One of the few rigorous analytical results for the time-varying frequency estimation was given by Huang and Hannan (1993). A recursive estimator with a data-driven sample size was studied. The iterated logarithm convergence rate $O((\log \log N/N^3)^{1/2})$ was obtained for a slowly time-varying frequency.

We want to emphasize that the slowly time-varying frequency hypothesis, which means that the first derivative has to be small, is a crucial one for most of the developed methods, except maybe the maximum likelihood (ML) estimator of the polynomial phase (Peleg and Porat, 1991).

This paper presents a new nonparametric approach to the estimation of the rapidly time-varying frequency. The nonparametric high-order local polynomial approximation (LPA) is used to design estimation algorithms.

The LPA as a locally weighted moving regression, appeared in statistical literature in Stone (1977), Cleveland (1979), Katkovnik (1979), (see also Katkovnik, 1985, Hardle, 1990). Earlier these ideas were applied in time-series analyses (e.g. Brown, 1963, and Box and Jenkins, 1970).

The results presented in this paper are a development of those given by Katkovnik (1994a,b), where the LPA was used for estimating time-varying phase, frequency and amplitude for the harmonic $A(t)\sin(\varphi(t))$. In those papers it was shown that the bias of estimation errors is proportional to the derivatives of the parameters in question with orders of the derivatives which depend on the orders of the LPA. The asymptotic MSE was derived and the estimates were shown to be asymptotically Gaussian.

The structure of the paper is as follows. The setting of the problem is presented in section 2. In section 3 the local polynomial time-frequency transformation (LPTFT) and the local polynomial periodogram (LPP) are introduced. Section 4 contains a motivation of the new estimates. Asymptotic properties of the estimates including the strong convergence statement, covariance matrices and upper bounds for bias of the estimates are given in section 5. Simulation results are discussed in section 6. The formulas for the asymptotic covariance matrix and bias of the estimates are derived in the appendix.

The LPTFT and LPP were proposed by Katkovnik (1994d).

2. SETTING OF THE PROBLEM

The model considered is

$$(1) \quad \begin{aligned} y(t) &= m(t) + \varepsilon(t), \\ m(t) &= A \sin(\varphi(t)) + B \cos(\varphi(t)), \quad t = \dots, -2, -1, 0, 1, 2, \dots, \end{aligned}$$

where the $\{\varepsilon(t)\}$ are independently and identically distributed random variables with $E\varepsilon(t) = 0$, $E\varepsilon^2(t) = \sigma^2$. A , B , $\varphi(t)$ are unknown time-invariant amplitudes and time-varying phase.

The instantaneous frequency (IF) is defined by

$$(2) \quad \Omega(t) = d\varphi(t)/dt.$$

The a priori information about $\Omega(t)$ is assumed to be given by assumptions (H1) and (H2).

(H1). $\Omega(t)$ is an m -differentiable function and

$$(3) \quad |d^m \Omega / dt^m| \leq L_m$$

(H2). Upper and lower bounds for r -derivatives, $r < m$, are known:

$$(4) \quad |d^r \Omega / dt^r| \leq L_r, \quad r = 0, 1, 2, \dots, m-1.$$

(H1), the condition of the existence of the presented estimates, is the principal assumption about the unknown parameters. It can be shown that the estimation is efficient provided L_m is small.

(H2) contains an additional information about $\Omega(t)$, which can be used to improve the estimator.

The $\Omega(t)$ is to be estimated from measurements $\{y(t)\}$.

3. LOCAL POLYNOMIAL TIME-FREQUENCY TRANSFORM AND LOCAL POLYNOMIAL PERIODOGRAM

The local polynomial time-frequency transform (LPTFT) is introduced in this paper as follows.

$$(5) \quad F_h(\omega, t) = \sum_{u=-h}^h y(t+u) \exp(-j\theta(u, \omega)), \quad \omega \in R^m,$$

where

$$(6) \quad \theta(u, \omega) = \omega_1 \cdot u + \omega_2 \cdot u^2/2 + \dots + \omega_m \cdot u^m/m!, \quad j = \sqrt{-1},$$

and m is the degree of the LPTFT.

One can see that

$$(7) \quad y(t+u) = \lim_{M \rightarrow \infty} \frac{1}{(2M+1)^m} \left(\sum_{\omega_1=-M}^M \dots \sum_{\omega_m=-M}^M F_h(\omega, t) \cdot \exp(j\theta(u, \omega)) \right)$$

follows directly from the definition (5) of the LPTFT and formulas

$$(8) \quad \sum_{k=-M}^M e^{juk} = 2 \cos \frac{Mk}{2} \cdot \sin \frac{(M+1)k}{2} \cdot \operatorname{cosec} \frac{k}{2},$$

$$\lim_{M \rightarrow \infty} \frac{1}{2M+1} \sum_{k=-M}^M e^{juk} = \begin{cases} 1, & \text{if } k = 0, \\ 0, & \text{if } k \neq 0. \end{cases}$$

(5) and (7) define the pair: the direct and the inverse LPTFT of degree m .

The direct LPTFT maps a real signal of time, $y(t)$, into a complex function $F_h(\omega, t)$ of m variables $\omega \in R^m$ and time t .

The inverse LPTFT maps the complex function $F_h(\omega, t)$ of $(m+1)$ variables into the real time signal $y(t)$.

We call

$$(9) \quad I_h(\omega, t) = |F_h(\omega, t)|^2, \quad \omega \in R^m,$$

a local polynomial periodogram (LPP).

The following multivariable version of Parseval's theorem is true for the LPTFT

$$(10) \quad \lim_{M \rightarrow \infty} \frac{1}{(2M+1)^m} \sum_{\omega_1 = -M}^M \dots \sum_{\omega_m = -M}^M |F_h(\omega, t)|^2 = \sum_{u = -h}^h y^2(t+u).$$

Equation (10) is important because it shows that $F_h(\omega, t)$ can be treated as an energy distribution of $y(t)$ over $\omega \in R^m$ for time-instant t .

The estimator of the IF is introduced in this paper as follows:

$$(11) \quad \hat{\omega}(t, h) = \arg \max_{\omega \in Q \subset R^m} I_h(\omega, t),$$

$$Q = \{\omega: |\omega_{r+1}| \leq L_r, \quad r = 0, 1, \dots, m-1\},$$

where the L_r are given in (H2).

The components of $\hat{\omega}(t, h)$ give the following estimates: $\hat{\omega}_1(t, h)$ is an estimate for the frequency $\Omega(t)$ and $\hat{\omega}_{k+1}(t, h)$ are estimates for the derivatives $d^k \Omega(t)/dt^k$, $k = 1, 2, m-1$.

Both the convergence result, $\hat{\omega}(t, h) \rightarrow \omega^*(t)$ as $h \rightarrow \infty$, and the asymptotic accuracy of the estimator (11), which are presented later, show that the distribution $F_h(\omega, t)$ as a function of ω is concentrated in the point

$$(12) \quad \omega^*(t) = (\Omega(t), d\Omega(t)/dt, \dots, d^{m-1}\Omega(t)/dt^{m-1})^T \in R^m$$

The conventional tools of modern spectral analysis of nonstationary signals based on different linear and nonlinear time-frequency representations show what is the energy distribution in the t - Ω plane and are used to obtain estimates of $\Omega(t)$ (see e.g. Hlawatsch and Boudreaux-Bartels, 1992).

The LPTFT is a new linear time-frequency representation which shows a time-frequency energy distribution over the t - $(\Omega, d\Omega/dt, \dots, d^{m-1}\Omega/dt^{m-1})$ space. The concentration of $F_h(\omega, t)$ in the point $\omega^*(t)$ represents the principal motivation of the new transform.

If we are interested only in the frequency, the derivatives of the frequency can be treated as auxiliary variables of the problem and can be eliminated.

Call

$$(13) \quad MI_h(\omega_1, t) = \max_{\omega' \in Q' \in R^{m-1}} I_h(\omega', t),$$

where

$$(14) \quad Q' = \{\omega_r : |\omega_{r+1}| \leq L_r, r = 1, 2, \dots, m-1\}$$

a maximized LPP (Max-LPP).

The Max-LPP is a function of one variable, ω_1 , the maximizing of which can be applied to obtain an estimate of the IF.

The LPTFT proposed in this paper can be interpreted as a generalization of the usual short-time Fourier transform (STFT) (see e.g. Kay, 1988), which corresponds to the particular case of the LPTFT with $m = 1$ or $\omega_2 = \dots = \omega_m = 0$.

It can be seen in the next section that m determines the degree of a polynomial which is used in order to approximate the time-varying phase into the window $[t - h, t + h]$.

(11) and (12) show the meaning of the coefficients ω_r in the expansion (6) and explain how the a priori information (H2) can be involved in (11).

We want to emphasize that the aim of the high-order polynomial

approximation in (5) with $m > 1$ is firstly to obtain a high accuracy of estimation of $\Omega(t)$, with the estimates of the derivatives as a secondary result, and finally, it enables one to use the a priori information given in the form (H2).

The LPTFT and LPP were proposed on the basis of the LPA approach. The relation to the LPA is of importance in order to understand the concept behind the new transform.

4. LOCAL POLYNOMIAL APPROXIMATION

The main idea of the LPA approach is to apply a truncated Taylor series in order to approximate time-varying parameters. Firstly, a part of a standard series is used to approximate the time-varying frequency and secondly this expansion is exploited locally to approximate the frequency on a small time-interval only. In fact, the local expansion is used to calculate an estimate for a single time-instant. For the next time-instant the calculations should be repeated. This determines the nonparametric character of this point-wise estimation. The localization of the estimation is ensured on the basis of the moving LS method with a weight-function (window) which discounts observations outside a neighbourhood of the centre of the approximation. More details about the LPA can be found in Katkovnik (1979, 1985, 1993, 1994c).

The point-wise estimation ensures the reproduction properties with respect to the polynomial components of $\Omega(t)$. It should be emphasized that the estimate of $\Omega(t)$ is not assumed to be a polynomial function of time. This is the principal difference between the nonparametric LPA and the maximum likelihood estimator of the polynomial phase (Peleg and Porat, 1991) where the constant coefficients of a polynomial series for the phase are estimated.

For the model (1) and the LPTFT (5) the loss function of the LPA (Katkovnik, 1994a, b) is of the form

$$(15) \quad J_h = \sum_{u=-h}^h (y(t+u) - [C_1 \sin \theta(u, \omega) + C_2 \cos \theta(u, \omega)])^2,$$

where $\theta(u, \omega)$ is given by (6).

The high-order LPA estimates of $\Omega(t)$, C_1 and C_2 have been introduced as a solution of the following constrained optimization problem:

$$(16) \quad (\tilde{C}_1, \tilde{C}_2, \tilde{\omega}) = \arg \min_{\omega \in Q, C_1, C_2} J_h.$$

$\tilde{\omega}$ gives the estimate of the $\omega^*(t)$ (12).

This estimator is justified as follows. The Taylor series for $\varphi(t+u)$ about the point t is of the form:

$$(17) \quad \varphi(t+u) = \varphi(t) + \theta(u, \Omega) + \Delta\varphi(t', u),$$

where

$$(18) \quad \theta(u, \Omega) = \Omega \cdot u + d\Omega/dt \cdot u^2/2 + \dots + d^{m-1}\Omega/dt^{m-1} \cdot u^m/m!,$$

$$\Delta\varphi(t', u) = d^m\Omega(t')/dt^m \cdot u^{m+1}/(m+1)!$$

and t' is the usual interior value of the argument, $t' = t + \lambda \cdot u$, $0 \leq \lambda \leq 1$.

Let $e(t+u)$ be a residual of the loss function (15) and $\varepsilon(t+u) = 0$. Substituting (17) one can obtain

$$(19) \quad e(t+u) = m(t+u) - [C_1 \sin(\theta(u, \omega)) + C_2 \cos(\theta(u, \omega))] = C_1^* \sin(\theta(u, \Omega) + \Delta\varphi(t', u)) + C_2^* \cos(\theta(u, \Omega) + \Delta\varphi(t', u)) - C_1 \sin(\theta(u, \omega)) + C_2 \cos(\theta(u, \omega)),$$

where

$$(20) \quad C_1^* = A \cos \varphi(t) - B \sin \varphi(t), \quad C_2^* = A \sin \varphi(t) + B \cos \varphi(t).$$

Let $\Delta\varphi = 0$ and let identity $e(t+u) = 0$ hold for all u . Then it can be seen that

$$(21) \quad \tilde{C}_1 = C_1^*, \quad \tilde{C}_2 = C_2^*, \quad \tilde{\omega} = \omega^*.$$

(21) shows that the components of ω can be interpreted as estimators of the time-varying frequency and its derivatives. The remainder term of the Taylor series and random errors $\varepsilon(t)$ result in estimation errors, i.e. in bias and variance of the estimates.

The usual arguments (e.g. Walker, 1971) can be used to derive (9) and

(11) from (15) and (16). Let us modify the definition of the LPA estimator slightly by replacing J_h (15) by the expression

$$(22) \quad U_h = \sum_{u=-h}^h \{y^2(t+u) - 2[C_1 \sin \theta(u, \omega) + C_2 \cos \theta(u, \omega)]y(t+u) + \frac{1}{2}(C_1^2 + C_2^2)\}.$$

This appreciably simplifies the estimation equation, and if in addition

$$(23) \quad J_h - U_h = \frac{1}{2} \sum_{u=-h}^h \{(A^2 - B^2) \cos(2\varphi(t+u)) + 2AB \sin(2\varphi(t+u))\}$$

is $O(1)$ as $h \rightarrow \infty$, we may expect the effect of this simplification on the estimators to be negligible for large h . We denote the modified estimates by

$\hat{C}_1, \hat{C}_2, \hat{\omega}$ respectively. Then it can be shown that

$$(24) \quad \begin{aligned} \hat{C}_1 &= \frac{2}{2h+1} \sum_{u=-h}^h y(t+u) \sin \theta(u, \hat{\omega}), \\ \hat{C}_2 &= \frac{2}{2h+1} \sum_{u=-h}^h y(t+u) \cos \theta(u, \hat{\omega}), \end{aligned}$$

where $\hat{\omega}$ is such that

$$(25) \quad I_h(\hat{\omega}_h) = \max_{\omega \in Q} I_h(\omega),$$

and I_h is given by (9). If (23) is negligible, then $\hat{\omega}_h \approx \tilde{\omega}_h$ as $h \rightarrow \infty$ and the optimization problems (11) and (25) are asymptotically equivalent to each other.

5. CONVERGENCE AND ASYMPTOTIC PROPERTIES OF THE ESTIMATE

To consider asymptotic properties the following assumptions are made.

(H3). $\varphi(t)$ is a polynomial of the degree less or equal to m ($L_m = 0$ in (H1)).

(H4). $\omega^*(t)$ is an interior point of Q and

$$(26) \quad S_r(h) = \frac{1}{h^{r+1}} \sum_{u=-h}^h \sin(\theta(u, \omega^*(t))) \cdot u^r = O(1/h),$$

$$C_r(h) = \frac{1}{h^{r+1}} \sum_{u=-h}^h \cos(\theta(u, \omega^*(t))) \cdot u^r = O(1/h), \text{ as } h \rightarrow \infty,$$

where r is an integer and $O(h^{-1})$ means that $O(h^{-1})/h^{-1} \rightarrow \text{constant}$ as $h \rightarrow \infty$.

PROPOSITION 1. Let $\hat{\omega}(t, h)$ be a solution of (11), Q be bounded and $\varepsilon(t)$ in (1) have the Gaussian distribution. Under assumptions (H3) to (H4) the estimators $\hat{\omega}(t, h)$ are strongly consistent for $\omega^*(t)$ as $h \rightarrow \infty$, i.e.

$$\omega(t, h) \xrightarrow{a.s.} \omega^*(t), \text{ as } h \rightarrow \infty. \quad \square$$

The proof of proposition 1 is quite similar to the proof which is given in Katkovnik (1994a, b) for the strong convergence of the LPA estimator of time-varying frequency and amplitude. The technique used is a modification of one given by Wu (1981).

Denote:

$$(27) \quad D(h) = \text{diag}(2h+1, (2h+1)^2, \dots, (2h+1)^m),$$

$$(28) \quad U = (u, u^2/2, \dots, u^m/m!)^T \in R^m,$$

$$(29) \quad W_\omega = \lim_{h \rightarrow \infty} \frac{1}{2h+1} D^{-1}(h) \left(\sum_{u=-h}^h U \cdot U^T - \frac{1}{2h+1} \sum_{u=-h}^h U \cdot \sum_{u=-h}^h U^T \right) D^{-1}(h),$$

$$(30) \quad b = \frac{2}{(m+1)!} \lim_{h \rightarrow \infty} \frac{1}{(2h+1)^{m+2}} D^{-1}(h) \left[\sum_{u=-h}^h |U| \cdot |u|^{m+1} + \frac{1}{2h+1} \sum_{u=-h}^h |U| \cdot \sum_{u=-h}^h |u|^{m+1} \right],$$

where $|U|$ is a vector of absolute values of U .

PROPOSITION 2. Let $\hat{\omega}(t, h)$ be a solution of (11), Q be bounded and $\epsilon(t)$ in (1) have Gaussian distribution.

Let (H1), (H2), (H4) hold and as $h \rightarrow \infty$

$$(31) \quad L_m \cdot h^m \rightarrow 0,$$

then

$D(h)(\hat{\omega}(t, h) - \omega^*(t))$ converges in law to

$$(32) \quad N\left(b(h), \frac{2\sigma^2}{A^2 + B^2} W_\omega^{-1} \frac{1}{2h+1}\right),$$

where

$$(33) \quad |b(h)| < L_m \cdot |W_\omega^{-1}| \cdot (2h+1)^{m+1} \cdot b + O(h^1).$$

Here $|W_\omega^{-1}|$ is a matrix of absolute values of W_ω^{-1} . Inequality $|x| \leq |y|$, $x, y \in R^m$, means that $|x_j| \leq |y_j|$, $j = 1, 2, \dots, m$. $\square$

The main elements of the proof of proposition 2 are given in the appendix.

It follows from proposition 2 that the asymptotic MSE is of the form:

$$(34) \quad E(d^{k-1}\Omega(t)/dt^{k-1} - \hat{\omega}_k(t))^2 = O\left(\left[\frac{\sigma^2 \cdot h^{-(1+2k)}}{A^2 + B^2}\right] + L_m \cdot h^{2(m-k+1)}\right),$$

where $k = 1, 2, \dots, m$.

The term $O(1)$ is important in (33) when $L_m = 0$. But it can be seen that this term can be omitted in (34) because it approaches zero faster than the standard deviation of $\Delta\omega$.

Proposition 2 and (34) give a clear picture of the estimation accuracy. The constant for MSE (34) can be accurately calculated from (29), (30) and (33).

6. SIMULATION

A large number of simulations have been done to study the LPP as an estimator of the IF and its derivatives. Here we wish to demonstrate only that the LPP has energy concentration in the point $\omega^*(t)$ (noiseless case).

The following model for $\varphi(t)$ is considered:

$$\varphi(t) = a_1 \cdot t + a_2 \cdot t^2/2, \quad \Omega(t) = a_1 + a_2 \cdot t, \quad d\Omega(t)/dt = a_2.$$

Figures 1, a, b, c show $I_h(\omega, t)$, $\omega = (\omega_1, \omega_2)^T$, $m = 2$, with parameters: $a_1 = 2\pi$, $a_2 = 8\pi$, $T = 0.1s$, $h = 20, 50$ and 100 . The four peaks in the figures correspond to the four time-instants $t = (0, 0.3, 0.6, 1.0)$ with IF values $\Omega(t) = (6.28, 13.82, 21.36, 31.42)s^{-1}$ and $d\Omega(t)/dt = 25.13s^{-2}$. The concentration of $I_h(\omega, t)$ in these four points $(\Omega(t), d\Omega/dt)$ is clear. The peaks of $I_h(\omega, t)$ become sharper if h increases.

Figures 2, a, b, c show the Max-LPP, $MI_h(\omega_1, t) = \max_{\omega_2} I_h(\omega_1, \omega_2; t)$, as a function of ω_1 for the same four time-instants. The straight line in the $t - \Omega(t)$ plane is the $\Omega(t)$ function. The linear IF law can be easily recovered from the peaks of the Max-LPP.

Figures 1 and 2 illustrate that the peaks of the LPP and the Max-LPP, i.e. the energy concentrations of the corresponding distributions, locate in the neighbourhood of the true values of the IF and its first derivative.

7. CONCLUSIONS

A new form of the time-frequency transform LPTFT and a new approach to the estimation of the rapidly time-varying frequency of a harmonic signal are proposed and justified. The most important distinctive feature of the new transform is that the degree of the complex exponential function generic for the FT is a polynomial function in the LPTFT.

The LPTFT is applied in order to propose a new form of the periodogram LPP where the argument is a vector whose components provide estimates of the IF and its derivatives.

The convergence results and asymptotic MSE for the estimator give an insight into properties of the LPP and LPTFT.

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APPENDIX

The asymptotic covariance matrix (32) and the upper bounds (33) for the bias of the estimates are derived here. Mathematical details of the limit passages and the proof of the asymptotic Gaussian distribution are omitted.

1. The strong convergence result (proposition 1) means that $\hat{\omega}(t, h) \rightarrow \omega^*(t)$ as $h \rightarrow \infty$, i.e. almost surely every $\hat{\omega}(t, h)$ eventually takes its values in a convex neighbourhood of $\omega^*(t)$ which is interior to Q because of (H4) and (31).

Since U_h defined by (22) is a minimum when $C_1 = \hat{C}_1$, $C_2 = \hat{C}_2$, $\omega = \hat{\omega}$ and $\hat{C}_1, \hat{C}_2, \hat{\omega}$ are interior to Q , then

$$(A.1) \quad \frac{\partial U_h(\hat{C}_1, \hat{C}_2, \hat{\omega})}{\partial \omega} = 0, \quad \frac{\partial U_h(\hat{C}_1, \hat{C}_2, \hat{\omega})}{\partial C_i} = 0, \quad k = 1, 2.$$

Substituting (17) gives (22) in the form:

$$(A.2) \quad U_h = \sum_{u=-h}^h \{y^2(t+u) - 2[C_1 \sin \theta(u, \omega) + C_2 \cos \theta(u, \omega)] \cdot y(t+u) + \frac{1}{2}(C_1^2 + C_2^2)\},$$

$$y(t+u) = C_1^*(t) \sin(\theta(u, \Omega) + \Delta \varphi(t', u)) + C_2^*(t) \cos(\theta(u, \Omega) + \Delta \varphi(t', u)),$$

where the C_i^* are given by (20).

It is easy to obtain

$$(A.3) \quad \begin{aligned} \partial U_h / \partial \omega &= -2 \sum_{u=-h}^h [C_1 \cos(\theta(u, \omega)) - C_2 \sin(\theta(u, \omega))] \cdot y(t+u) \cdot U, \\ \partial U_h / \partial C_1 &= (2h+1)C_1 - 2 \sum_{u=-h}^h \sin(\theta(u, \omega)) \cdot y(t+u), \\ \partial U_h / \partial C_2 &= (2h+1)C_2 - 2 \sum_{u=-h}^h \cos(\theta(u, \omega)) \cdot y(t+u). \end{aligned}$$

Here and in what follows, according to (28),

$$U = (u, u^2/2, \dots, u^m/m!)^T \in R^m.$$

Let us assume that the values

$$(A.4) \quad \Delta \hat{C}_1 = C_1^* - \hat{C}_1, \quad \Delta C_2 = C_2^* - \hat{C}_2, \quad \Delta \omega = \omega^* - \hat{\omega}, \quad \Delta \varphi, n$$

are small so that the derivatives in (A.1) can be approximated by the first terms of the Taylor series with respect to these values.

For (A.1) and (A.3) the corresponding expansions give

$$(A.5) \quad \begin{aligned} (U_h)_{\hat{\omega}} &= (U_h)_{\omega^*} - (U_h)_{\omega^* \omega^{*T}} \Delta \omega - (U_h)_{\omega^* C_1^*} \Delta C_1 - (U_h)_{\omega^* C_2^*} \Delta C_2 + \\ &\quad \delta_\varphi(U_h)_{\omega^*} + \delta_n(U_h)_{\omega^*}, \\ (U_h)_{\hat{C}_1} &= (U_h)_{C_1^*} - (U_h)_{C_1^* \omega^{*T}} \Delta \omega - (U_h)_{C_1^* C_1^*} \Delta C_1 - (U_h)_{C_1^* C_2^*} \Delta C_2 + \\ &\quad \delta_\varphi(U_h)_{C_1^*} + \delta_n(U_h)_{C_1^*}, \\ (U_h)_{\hat{C}_2} &= (U_h)_{C_2^*} - (U_h)_{C_2^* \omega^{*T}} \Delta \omega - (U_h)_{C_2^* C_1^*} \Delta C_1 - (U_h)_{C_2^* C_2^*} \Delta C_2 + \\ &\quad \delta_\varphi(U_h)_{C_2^*} + \delta_n(U_h)_{C_2^*} \end{aligned}$$

where

$$(U_h)_{\hat{\omega}} \text{ denotes } \partial U_h(\hat{C}_1, \hat{C}_2, \hat{\omega}, \Delta \varphi, n) / \partial \omega,$$

$$(U_h)_{\hat{C}_i} \text{ denotes } \partial U_h(\hat{C}_1, \hat{C}_2, \hat{\omega}, \Delta \varphi, n) / \partial C_i,$$

$$(U_h)_{\omega^*} \text{ denotes } \partial U_h(C_1^*, C_2^*, \omega^*, 0, 0) / \partial \omega,$$

$$(U_h)_{C_i^*} \text{ denotes } \partial U_h(C_1^*, C_2^*, \omega^*, 0, 0) / \partial C_i,$$

$$(U_h)_{\omega^* \omega^{*T}} \text{ denotes } \partial^2 U_h(C_1^*, C_2^*, \omega^*, 0, 0) / \partial \omega \partial \omega^T,$$

$$(U_h)_{\omega^* C_i^*} \text{ denotes } \partial^2 U_h(C_1^*, C_2^*, \omega^*, 0, 0) / \partial \omega \partial C_i, \text{ etc.}$$

We point out that $\Delta \varphi$ and n are equal to zero in all of the derivatives with asterisks.

The $\delta_{\varphi}(\cdot)$ and $\delta_n(\cdot)$ in (A.5) denote linearized increments of the derivatives $(U_h)_{\omega^*}^*$, $(U_h)_{C_i^*}^*$, which correspond to $\Delta\varphi$ and n respectively:

$$(A.6) \quad \delta_{\varphi}(U_h)_{\omega^*}^* = -2 \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))]^2 U \Delta\varphi(t, u),$$

$$(A.7) \quad \delta_n(U_h)_{\omega^*}^* = -2 \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))] U n(t+u)$$

$$(A.8) \quad \delta_{\varphi}(U_h)_{C_1^*}^* = -2 \sum_{u=-h}^h \sin(\theta(u, \omega^*)) [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))] \Delta\varphi(t, u),$$

$$(A.9) \quad \delta_n(U_h)_{C_1^*}^* = -2 \sum_{u=-h}^h \sin(\theta(u, \omega^*)) n(t+u),$$

$$(A.10) \quad \delta_{\varphi}(U_h)_{C_2^*}^* = -2 \sum_{u=-h}^h \cos(\theta(u, \omega^*)) [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))] \Delta\varphi(t, u),$$

$$(A.11) \quad \delta_n(U_h)_{C_2^*}^* = -2 \sum_{u=-h}^h \cos(\theta(u, \omega^*)) n(t, u).$$

It follows from (A.5) to (A.11) that the following linear equation determines $\Delta\omega$ and ΔC_i :

$$(A.12) \quad \Phi_h \begin{bmatrix} \Delta \omega \\ \Delta C_1 \\ \Delta C_2 \end{bmatrix} = \tilde{n} + B_1(h) + B_2(h),$$

where

$$(A.13) \quad \Phi_h = \begin{bmatrix} (U_h)_{\omega^* \omega^* T} & (U_h)_{\omega^* C_1^*} & (U_h)_{\omega^* C_2^*} \\ (U_h)_{C_1^* \omega^* T} & (U_h)_{C_1^* C_1^*} & (U_h)_{C_1^* C_2^*} \\ (U_h)_{C_2^* \omega^* T} & (U_h)_{C_2^* C_1^*} & (U_h)_{C_2^* C_2^*} \end{bmatrix},$$

$$(A.14) \quad \tilde{n} = (\delta_n(U_h)_{\omega^*}^T, \delta_n(U_h)_{C_1^*}^T, \delta_n(U_h)_{C_2^*}^T)^T,$$

$$(A.15) \quad B_1(h) = (\delta_\varphi(U_h)_{\omega^*}^T, \delta_\varphi(U_h)_{C_1^*}^T, \delta_\varphi(U_h)_{C_2^*}^T)^T.$$

$$B_2(h) = ((U_h)_{\omega^*}^T, (U_h)_{C_1^*}^T, (U_h)_{C_2^*}^T)^T.$$

The elements of Φ_h are of the form:

(A.16)

$$(U_h)_{\omega^* \omega^* T} = 2 \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) + C_2^* \sin(\theta(u, \omega^*))]^2 U U^T,$$

$$(U_h)_{\omega^* C_1^*} = -2 \sum_{u=-h}^h \cos(\theta(u, \omega^*)) (C_1^* \sin(\theta(u, \omega^*)) + C_2^* \cos(\theta(u, \omega^*))) U$$

$$(U_h)_{\omega^* C_2^*} = 2 \sum_{u=-h}^h \sin(\theta(u, \omega^*)) (C_1^* \sin(\theta(u, \omega^*)) + C_2^* \cos(\theta(u, \omega^*))) U,$$

$$(U_h)_{C_1^* C_1^*} = 2h + 1,$$

$$(U_h)_{C_1^* C_2^*} = 0,$$

$$(U_h)_{C_2^* C_2^*} = 2h + 1.$$

The elements of $B_1(h)$ and $\tilde{n}$ are given by (A.5) to (A.11) according to (A.14) to (A.15).

2. The covariance of the estimates is evaluated from (A.12):

$$(A.17) \quad \text{cov}((\Delta\omega, \Delta C_1, \Delta C_2)^T) = \sigma^2 \cdot \Phi_h^{-1} \cdot \text{cov}(\tilde{n}) \cdot \Phi_h^{-1}$$

where the elements of $\text{cov}(\tilde{n}) = E\tilde{n}\tilde{n}^T/\sigma^2$ are as follows:

(A.18)

$$\text{cov}(\tilde{n})_{11} = 4 \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))]^2 U U^T,$$

$$\text{cov}(\tilde{n})_{12} = 4 \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))] U \sin(\theta(u, \omega^*))$$

$$\text{cov}(\tilde{n})_{13} = 4 \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))] U \cos(\theta(u, \omega^*)),$$

$$\text{cov}(\tilde{n})_{22} = 4 \sum_{u=-h}^h \sin^2(\theta(u, \omega^*)),$$

$$\text{cov}(\tilde{n})_{23} = 2 \sum_{u=-h}^h \sin(2\theta(u, \omega^*))$$

$$\text{cov}(\tilde{n})_{33} = 4 \sum_{u=-h}^h \cos^2(\theta(u, \omega^*)).$$

For the bias one can obtain from (A.12):

$$(A.19) \quad E(\Delta\omega, \Delta C_1, \Delta C_2)^T = \Phi_h^{-1} \cdot (B_1(h) + B_2(h)).$$

The $B_1(h)$ and $B_2(h)$ represent two additive components of the bias which have a different origin. The bias corresponding to $B_1(h)$ is determined by the properties of the estimated function $\Omega(t)$. It is equal to 0, if $\varphi(t)$ is a polynomial with degree equal or less than m . This bias depends on the difference between $\varphi(t)$ and the polynomial function.

It is well known for the time-invariant frequency that the usual short-time Fourier periodogram is able to give only an approximation of the true frequency even if we have no noise in the observations. This approximating frequency value converges to the true frequency when the number of observations increases. The second component of the bias

corresponding to $B_2(h)$ reflects this principal approximating property of the periodogram. This bias for the periodogram estimation of time-invariant frequency is discussed by Rice and Rosenblatt (1988).

Note that only the first kind of bias exists in the LPA frequency estimates (see Katkovnik, 1994a,c). As a matter of fact it means that a second kind of the bias appears as a result of the replacement of the LPA performance (15) by the simplified performance (22).

Denote these two components of the bias as follows:

$$(A.20) \quad E(\Delta\omega^{(k)}, \Delta C_1^{(k)}, \Delta C_2^{(k)})^T = \Phi_h^{-1} \cdot B_k(h), \quad k = 1, 2.$$

3. In what follows the asymptotic formulas for the covariance and the bias are derived for $h \rightarrow \infty$.

It follows from (24) that

$$(A.21) \quad \sum_{u=-h}^h \sin(\theta(u, \omega^*)) \cdot u^r = O(h^r),$$

$$\sum_{u=-h}^h \cos(\theta(u, \omega^*)) \cdot u^r = O(h^r)$$

where $O(h^r)$ means that $O(h^r)/h^r \rightarrow \text{constant}$ as $h \rightarrow \infty$.

For $(U_h)_{\omega^* \omega^* T}$ we obtain from (A.3)

$$(A.22) \quad (U_h)_{\omega^* \omega^* T} = \sum_{u=-h}^h \{ [(C_1^*)^2 + (C_2^*)^2] UU^T + C_1^* C_2^* \sin(2\theta(u, \omega^*)) - (C_2^*)^2 \cos(2\theta(u, \omega^*)) + (C_1^*)^2 \cos(2\theta(u, \omega^*)) UU^T \},$$

Now because of (A.21), every term of the sum (A.22), which is multiplied by harmonics with frequency $\theta(u, \Omega)$ has an order which is less than the order of the corresponding term without such a factor.

Then

$$(U_h)_{\omega^* \omega^* T} \rightarrow [(C_1^*)^2 + (C_2^*)^2] \sum_{u=-h}^h UU^T \text{ as } h \rightarrow \infty.$$

The similar analysis of the other items Φ_h and $\text{cov}(\tilde{n})$ shows that

$$(A.23) \quad \lim_{h \rightarrow \infty} \Phi_h = \begin{pmatrix} \Sigma_{u=-h}^h [(C_1^*)^2 + (C_2^*)^2] UU^T, & -C_2^* \Sigma_{u=-h}^h U & C_1^* \Sigma_{u=-h}^h U \\ -C_2^* \Sigma_{u=-h}^h U^T, & 2h+1 & 0 \\ C_1^* \Sigma_{u=-h}^h U^T, & 0, & 2h+1 \end{pmatrix}$$

and

$$(A.24) \quad \lim_{h \rightarrow \infty} \text{cov}(\tilde{n}) = 2 \cdot \lim_{h \rightarrow \infty} \Phi_h.$$

Then (A.17) is simplified,

$$(A.25) \quad \lim_{h \rightarrow \infty} \text{cov}(\Delta\omega, \Delta C_1, \Delta C_2)^T = 2\sigma^2 \cdot \lim_{h \rightarrow \infty} \Phi_h^{-1}.$$

(A.25) gives the covariance for the frequency and amplitude estimates. In order to find the covariance for frequency only the corresponding block of Φ_h^{-1} has to be extracted. In order to do it use the formula for an inverse block-matrix:

$$(A.26) \quad \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^{-1} = \begin{pmatrix} A_{11 \cdot 2}^{-1}, & -A_{11 \cdot 2}^{-1} \cdot A_{12} \cdot A_{22}^{-1} \\ -A_{22}^{-1} \cdot A_{21} \cdot A_{11 \cdot 2}^{-1}, & A_{22}^{-1} \end{pmatrix},$$

$$A_{11 \cdot 2} = A_{11} - A_{12} \cdot A_{22}^{-1} \cdot A_{21},$$

$$A_{22 \cdot 1} = A_{22} + A_{22}^{-1} \cdot A_{21} \cdot A_{11 \cdot 2}^{-1} \cdot A_{12} \cdot A_{22}^{-1}.$$

Let in (A.26)

$$(A.27) \quad \begin{aligned} A_{11} &= \Sigma_{u=-h}^h [(C_1^*)^2 + (C_2^*)^2] UU^T, \\ A_{12} &= A_{21}^T = (-C_2^* \Sigma_{u=-h}^h U, C_1^* \Sigma_{u=-h}^h U), \\ A_{22} &= (2h+1) \cdot \text{diag}(1, 1). \end{aligned}$$

It is clear that

$$(A.28) \quad A_{11 \cdot 2}^{-1} = \frac{1}{A^2 + B^2} \cdot F^{-1},$$

$$F = \sum_{u=-h}^h U U^T - \frac{1}{2h+1} \sum_{u=-h}^h U \cdot \sum_{u=-h}^h U^T$$

Then $\text{cov}(\Delta\omega) = 2\sigma^2 \cdot A_{11 \cdot 2}^{-1}$ and substituting (A.28) into (A.25) we obtain as $h \rightarrow \infty$

$$(A.29) \quad \text{cov}(\Delta\omega) = \frac{2\sigma^2}{A^2 + B^2} \cdot F^{-1}.$$

The formulas (A.28) and (A.29) give the covariance of the estimate $\Delta\omega$.

4. Consider the bias of $\Delta\omega^{(1)}$.

It follows from (A.20), (A.23) and (A.26) that

$$(A.30) \quad E(\Delta\omega^{(1)}) = A_{11 \cdot 2}^{-1} \cdot B_{\Delta\omega}(h) - A_{11 \cdot 2}^{-1} \cdot A_{12} \cdot A_{22}^{-1} \cdot B_{\Delta C}(h),$$

where, $B_{\Delta\omega}(h)$ and $B_{\Delta C}(h)$ are respectively the first m and the last two components of $B_1(h)$, $B_1(h) = \begin{bmatrix} B_{\Delta\omega}(h) \\ B_{\Delta C}(h) \end{bmatrix}$, $B_{\Delta\omega}(h) = \delta_\varphi(U_h)_{\omega^*}$, $B_{\Delta C}(h) = (\delta_\varphi(U_h)_{C_1^*}, \delta_\varphi(U_h)_{C_2^*})^T$. Substituting (A.21), (A.22), (A.6), (A.8) and

(A.10) into (A.30) one can see that

$$(A.31) \quad E(\Delta\omega^{(1)}) = -\frac{2}{A^2 + B^2} F^{-1} \cdot \left\{ \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))]^2 U \Delta\varphi(t, u) + \frac{1}{2h+1} \sum_{u=-h}^h U \cdot \sum_{u=-h}^h [C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*))]^2 \cdot \Delta\varphi(t, u) \right\}.$$

Upper bounds for the elements of $E(\Delta\omega^{(1)})$ can be easily obtained from (A.31) provided (3) and (26):

$$(A.32) \quad |E(\Delta\omega^{(1)})| \leq \frac{2L}{(m+1)!} \cdot |F^{-1}| \cdot \left\{ \sum_{u=-h}^h |U| |u|^{m+1} + \frac{1}{2h+1} \sum_{u=-h}^h |U| \cdot \sum_{u=-h}^h |u|^{m+1} \right\},$$

where the module of a vector or a matrix means a vector or a matrix of absolute values of the elements of the corresponding vector or matrix.

5. Consider the bias of $\Delta\omega^{(2)}$.

First, derive the orders of the elements of $B_2(h)$. It can be seen from (A.3) that

$$(A.33) \quad (U_h)_{\omega^*} = -2 \sum_{u=-h}^h \left[C_1^* \cos(\theta(u, \omega^*)) - C_2^* \sin(\theta(u, \omega^*)) \right] \times \\ \left[C_1^* \sin(\theta(u, \omega^*)) + C_2^* \cos(\theta(u, \omega^*)) \right] \cdot U = \\ -2 \sum_{u=-h}^h \left[(C_1^{*2} - C_2^{*2}) \frac{\sin(2\theta(u, \omega^*))}{2} + C_1^* C_2^* \cos(2\theta(u, \omega^*)) \right] \cdot U.$$

Then it follows from (A.21) and (A.33) that

$$(A.34) \quad (U_h)_{\omega^*} = (O(h), O(h^2), \dots, O(h^m))^T.$$

It can be shown in a similar way that

$$(A.35) \quad (U_h)_{C_1} = O(h), (U_h)_{C_2} = O(h).$$

The formula similar to (A.30) is of the form

$$(A.36) \quad E(\Delta\omega^{(2)}) = A_{11 \cdot 2}^{-1} \cdot (U_h)_{\omega^*} - A_{11 \cdot 2}^{-1} \cdot A_{12} \cdot A_{22}^{-1} \cdot [(U_h)_{C_1}, (U_h)_{C_2}]^T.$$

Substituting (A.26), (A.27), (A.34) and (A.35) into (A.36) gives

$$(A.37) \quad E(\Delta\omega^{(2)}) = F^{-1} \left[(O(h), O(h^2), \dots, O(h^m))^T + \frac{1}{2h+1} \sum_{u=-h}^h U \cdot O(h) \right].$$

6. The next steps enable one to represent the asymptotic relations of $\text{cov}(\Delta\omega)$ and $E(\Delta\omega)$ as functions of h in clear analytical form.

According to (27) and (29), let us introduce the diagonal matrix

$$(A.38) \quad D(h) = \text{diag}(2h+1, (2h+1)^2, \dots, (2h+1)^m)$$

and the matrix

$$(A.39) \quad W_{\omega} = \lim_{h \rightarrow \infty} \frac{1}{2h+1} D^{-1}(h) \left[\sum_{u=-h}^h U U^T - \frac{1}{2h+1} \sum_{u=-h}^h U \sum_{u=-h}^h U^T \right] D^{-1}(h).$$

It can be verified after substitution of (A.38) and (A.39) in (A.29) that

$$(A.40) \quad \text{cov}(D(h) \cdot \Delta \omega) = \frac{2\sigma^2}{(A^2 + B^2)} W_{\omega}^{-1} \frac{1}{2h+1}.$$

This formula gives the covariance matrix in (32).

The same substitution in (A.32) results in

$$(A.41) \quad |E(D(h) \cdot \Delta \omega^{(1)})| \leq L_m \cdot |W^{-1}| \cdot b \cdot (2h+1)^{m+1},$$

where b is given by (30):

$$(A.42) \quad b = \frac{2}{(m+1)!} \lim_{h \rightarrow \infty} \frac{1}{(2h+1)^{m+2}} D^{-1}(h) \left(\sum_{u=-h}^h |U|^{m+1} + \frac{1}{2h+1} \sum_{u=-h}^h |U| \cdot \sum_{u=-h}^h |u|^{m+1} \right).$$

In a similar way one can obtain for (A.37),

$$(A.43) \quad E(D(h) \cdot \Delta \omega^{(2)}) = \frac{1}{2h+1} W^{-1} D^{-1}(h) \left[(O(h), O(h^2), \dots, O(h^m))^T + \frac{1}{2h+1} \sum_{u=-h}^h U \cdot O(h) = O(1) \right].$$

Comparing (A.43) with (A.41), one sees that if $L_m \neq 0$ the bias $E(D(h) \cdot \Delta \omega^{(1)}) \rightarrow \infty$ while $E(D(h) \cdot \Delta \omega^{(2)}) \rightarrow O(1)$ as $h \rightarrow \infty$. The combination of (A.41) and (A.43) proves (33).

It is obvious that $L_m h^m$ should be small in order that the bias be small too, as it was assumed in the analysis which was done. In accurate terms it means (31).

$$(A.44) \quad L_m h^m \rightarrow 0 \text{ as } h \rightarrow \infty.$$

It completes the proof of (32) and (33) of proposition 2.

The more delicate conclusion of proposition 2 with respect to the Gaussian distribution of the estimates can be obtained using a technique developed in (Katkovnik, 1994a, b).

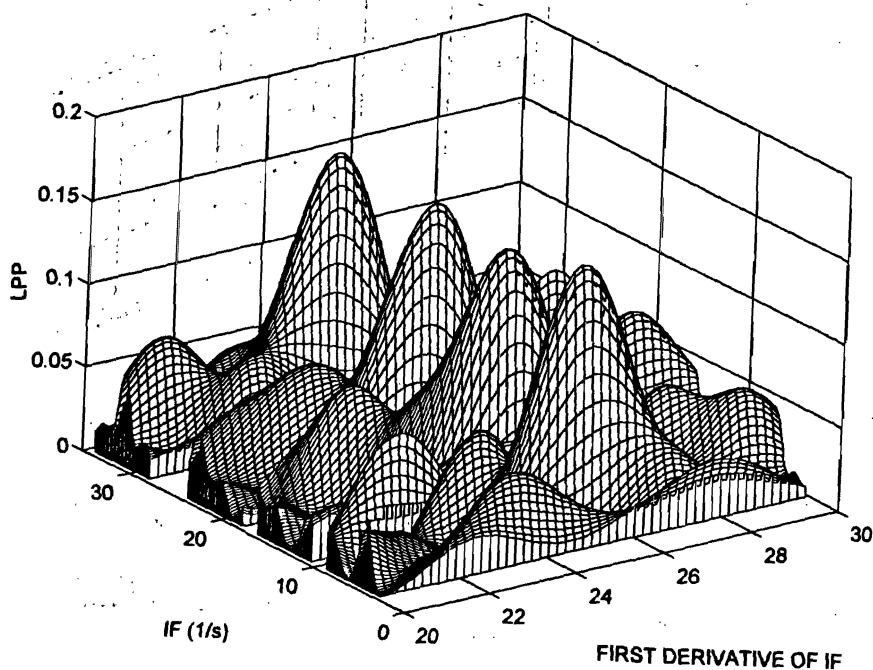


Figure 1a. Linear LPP versus IF (ω_1) and first derivative of IF (ω_2).
 The window size is $2h + 1$: $h = 20$;
 IF is $\Omega(t) = 2\pi + 8\pi t$, sampling frequency is 10 Hz.

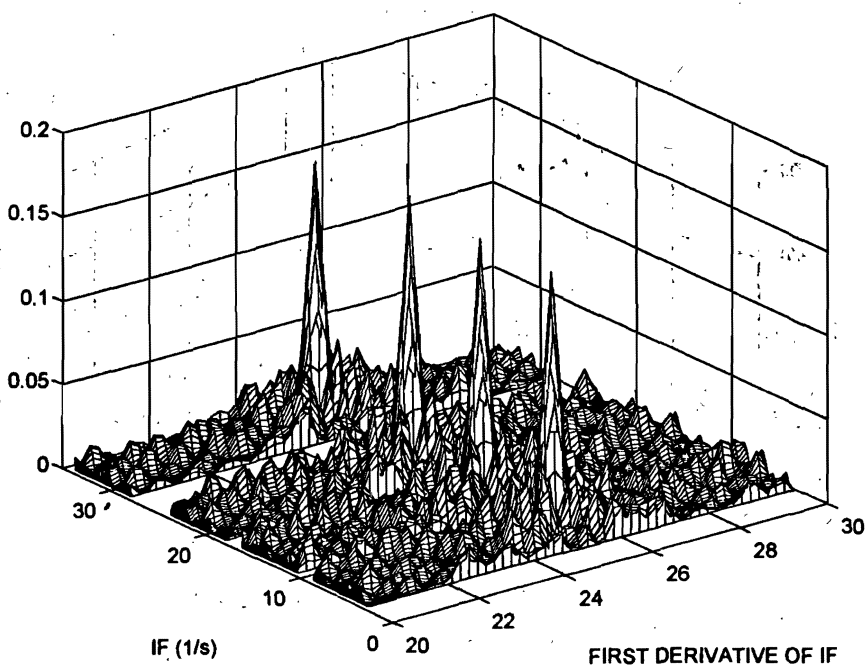


Figure 1b. Linear LPP versus IF (ω_1) and first derivative of IF (ω_2).
 The window size is $2h + 1$: $h = 50$;
 IF is $\Omega(t) = 2\pi + 8\pi t$, sampling frequency is 10 Hz.

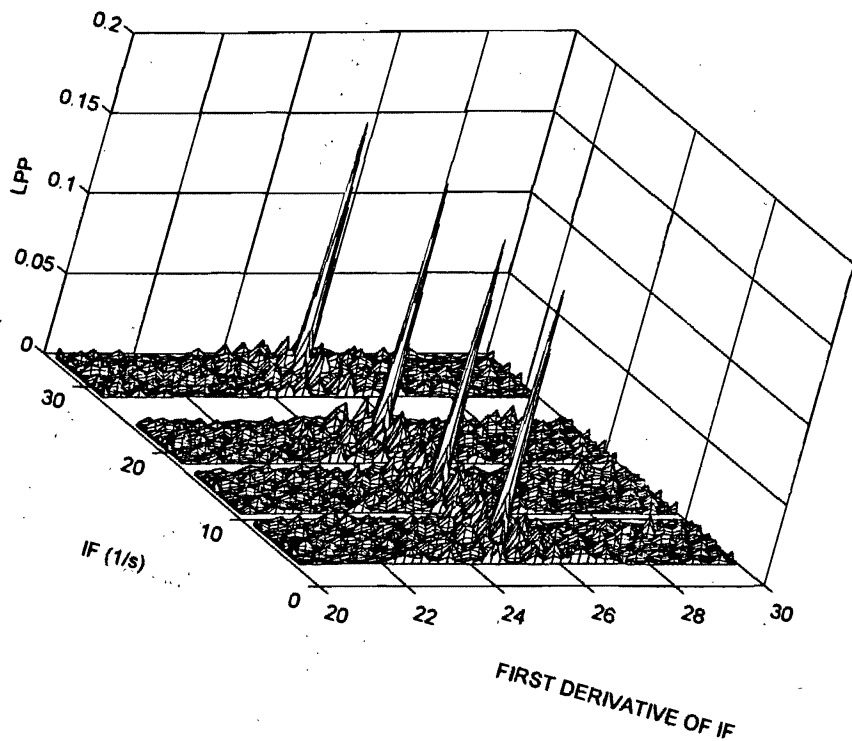


Fig. 1c.

Linear LPP versus IF (ω_1) and first derivative of IF (ω_2).
 The window size is $2h + 1$: $h = 100$;
 IF is $\Omega(t) = 2\pi + 8\pi t$, sampling frequency is 10 Hz.

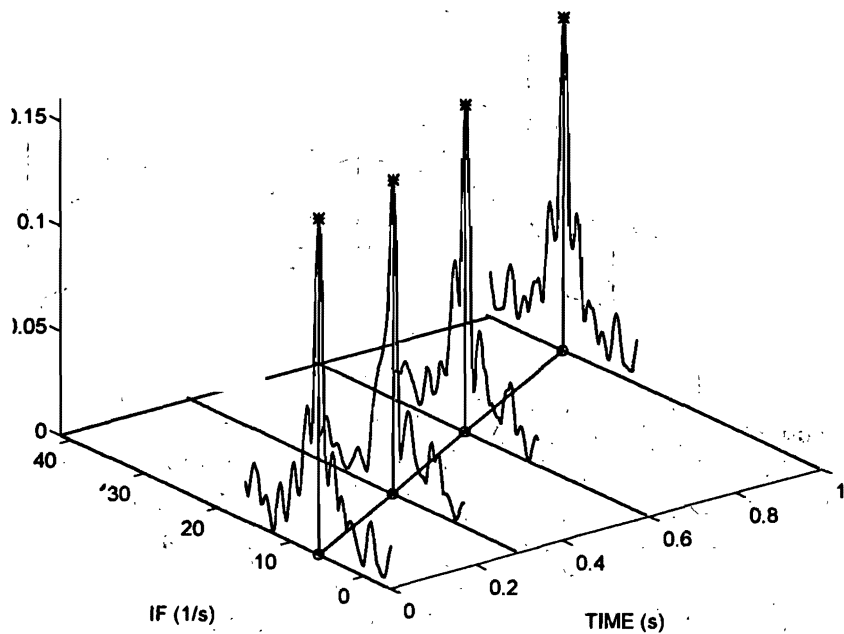


Figure 2a. Max-LPP versus IF (ω_1) and time.
 The window size is $2h + 1$: $h = 20$;
 IF is $\Omega(t) = 2\pi + 8\pi t$, $d\Omega(t)/dt = 8\pi$, sampling frequency is 10 Hz.

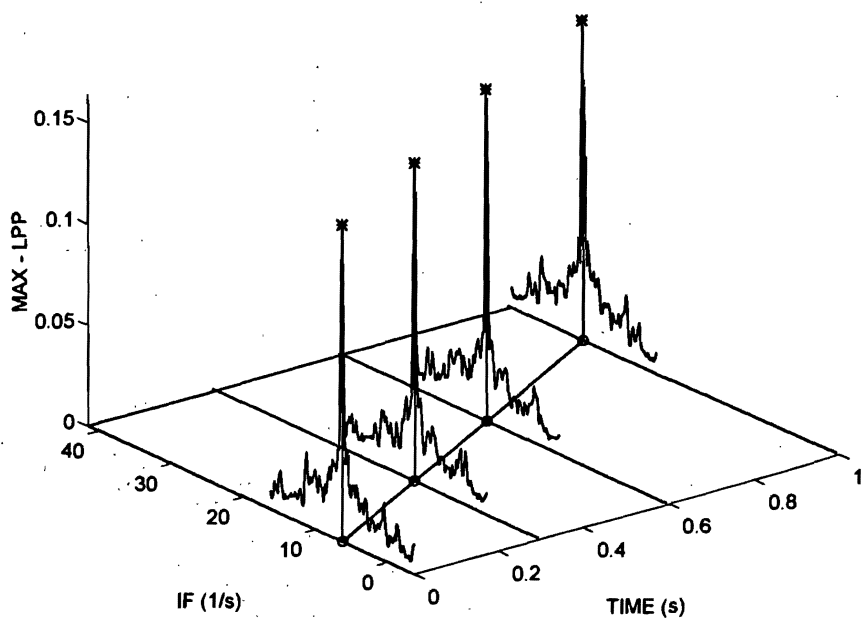


Figure 2b. Max-LPP versus IF (ω_1) and time
 The window size is $2h + 1$: $h = 50$;
 IF is $\Omega(t) = 2\pi + 8\pi t$, $\delta\Omega(t)/dt = 8\pi$, sampling frequency is 10 Hz.

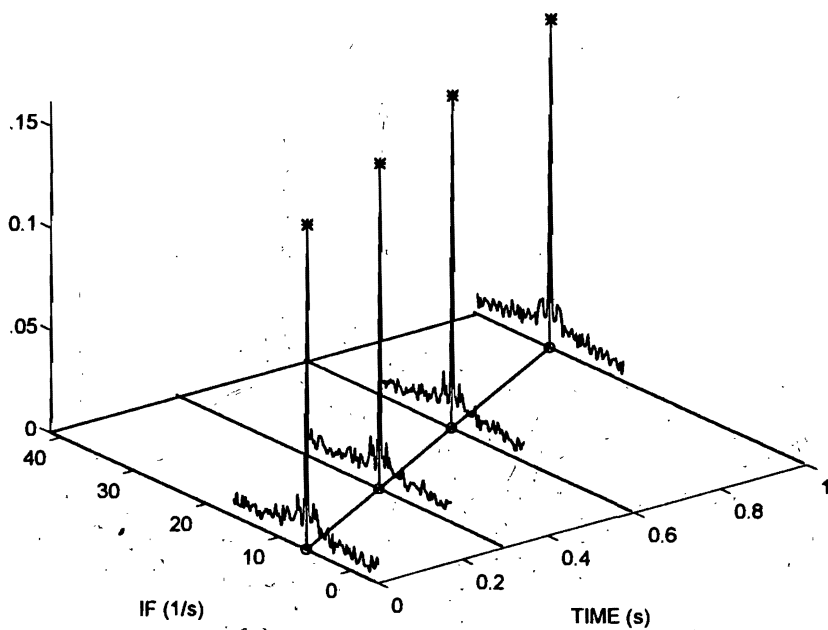


Figure 2c. Max-LPP versus IF (ω_1) and time.
 The window size is $2h + 1$: $h = 100$;
 IF is $\Omega(t) = 2\pi + 8\pi t$, $d\Omega(t)/dt = 8\pi$, sampling frequency is 10 Hz.